Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
MERN Stack

M1:Practical – I

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1.Environment Setup and JavaScript Basics:

a. Install Node.js and Visual Studio Code, and verify installation.

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL

PS C:\Users\itlab\Desktop\Asmitha\MERN STACK> node -v
v24.17.0
PS C:\Users\itlab\Desktop\Asmitha\MERN STACK> npm -v
11.13.0
PS C:\Users\itlab\Desktop\Asmitha\MERN STACK>
```

b. Create a simple Node.js file and run it using the terminal.

```
JS Prac-1b.js
console.log("Hello, Welcome to 1st practical of MERN");
console.log("Asmitha");
```

```
PS C:\Users\itlab\Desktop\Asmitha\MERN STACK> node Prac-1b.js
Hello, Welcome to 1st practical of MERN
Asmitha
PS C:\Users\itlab\Desktop\Asmitha\MERN STACK>
```

c. Write a Node.js script demonstrating basic JavaScript commands (console log, variables, and arithmetic operations).

```
JS prac-1c.js > ...
console.log("Welcome to Node.js!");
let num1 = 20;
let num2 = 10;
console.log("Add:", num1 + num2);
console.log("Sub:", num1 - num2);
console.log("Multiply:", num1 * num2);
console.log("Div:", num1/num2);
console.log("Asmitha")
```

```
PS C:\Users\itlab\Desktop\Asmitha\MERN STACK> node prac-1c.js
Welcome to Node.js!
Add: 30
Sub: 10
Multiply: 200
Div: 2
Asmitha
```

d. Create and run a simple program using variables and functions.

```
JS prac-1d.js > ...
let a = 12;
let b = 21;
function add(a,b){
    return a+b;
}
function multiply(a,b){
    return a*b;
}
console.log("ADD: ",add(a,b));
console.log("MULTIPLY: ",multiply(a,b));
console.log("Asmitha");
```

```
PS C:\Users\itlab\Desktop\Asmitha\MERN STACK> node prac-1d.js
ADD:  33
MULTIPLY:  252
Asmitha
```

e. Demonstrate use of operators and control statements in JavaScript

```
let a = 20;
let b = 10;
console.log("\nComparison Operators");
console.log("a > b",a>b);
console.log("a < b",a<b);
console.log("a == b",a==b);
console.log("a != b",a!=b);
console.log("\nIf-Else Statement");
if(a>b){
    console.log("a is greater than b");
}else{
    console.log("b is greater than or equal to a");
}
console.log("\nSwitch Statement");
let day = 2;
switch (day){
    case 1:
        console.log("Mon");
        break;
    case 2:
        console.log("Tue");
        break;
    case 3:
        console.log("Wed");
        break;
    default:
        console.log("Invalid");
}
console.log("Asmitha")
```

```
Comparison Operators
a > b true
a < b false
a == b false
a != b true

If-Else Statement
a is greater than b

Switch Statement
Tue
Asmitha
```

f. Write programs using loops and template literals.

```
JS prac-1f.js > ...
let num = 5;
console.log(`Multiplication Table of ${num}`);
for(let i = 1; i <= 10; i++)
{
    console.log(`${num} x ${i} = ${num * i}`);
}
```

```
Multiplication Table of 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```